

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

Analysis of Training and Testing MSE in a Neural Network

Given:

- Training Mean Squared Error (MSE) = 0.85
- Testing Mean Squared Error (MSE) = 5.20

Analysis:

The training MSE is much lower than the testing MSE. This indicates that the neural network performs very well on the training data but poorly on unseen test data. This is a sign of overfitting, where the model memorizes the training data instead of learning general patterns.

Possible Reasons:

1. Overfitting: The model is too complex and fits the training data too closely.
2. Insufficient Training Data: A small or unrepresentative dataset limits the model's ability to generalize.
3. Noise in Training Data: The model may have learned random noise rather than meaningful patterns.
4. Lack of Regularization: Techniques such as dropout or L2 regularization were not used.
5. Data Distribution Difference: The training and testing datasets may have different characteristics.

Corrective Measures:

- Use Dropout layers to reduce overfitting.
- Apply L1/L2 Regularization to penalize overly complex models.

- Increase the size and diversity of the training dataset.
- Use Early Stopping to stop training before overfitting occurs.
- Perform Cross-Validation to improve model evaluation.
- Simplify the neural network by reducing the number of hidden layers or neurons.
- Ensure proper data preprocessing and normalization.

Conclusion:

The large difference between the training MSE (0.85) and testing MSE (5.20) indicates that the model is overfitting. Applying regularization techniques, increasing training data, simplifying the model, and using early stopping can improve the model's ability to generalize and reduce the testing error.

Code of Conduct:

I .akhil(192525250)m certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch.akhil

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided